Hot Weather and Undernutrition in Sub-Saharan Africa: Vulnerabilities, Mechanisms, and Adaptation *

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Version 25 August, 2025

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Abstract

I study how extreme heat during the last growing season affects household nutrition in four Sub-Saharan African countries, a setting with a high prevalence of small family farms. Linking 19 waves of panel data from the World Bank Living Standards Measurement Study - Integrated Surveys on Agriculture (LSMS-ISA) panels (2008–2023) to high-resolution weather, I estimate within-household multi-way fixed effects models. High growing season temperatures reduce crop yields and daily energy availability, increasing the number of undernourished households, as measured by the consumption of calories, protein, iron, zinc, and folate. Proteins, which I demonstrate to be a key determinant of child stunting, are most affected. The main underlying pathway is likely income. Only wealthier households partially compensate for the reduced availability of own-grown food by purchasing more food. Extreme heat depresses total expenditure, reduces the share of agricultural output sold, shifts labour toward own-farm, and increases livestock losses. I investigate demand- and supply-driven price mechanisms as an additional pathway. Preliminary results suggest school feeding partly cushions nutritional losses. The findings highlight the need to pair on-farm adaptation with income smoothing and nutrition-focused safety nets.

Keywords: Agriculture; Climate change; Extreme weather; Nutrition; Sub-Saharan Africa.

JEL Codes: I15, O13, Q11, Q12, Q54.

*I thank Laure de Preux, Ariel Ortiz-Bobea, Franco Sassi, and participants to the Association of Environmental and Resource Economists 2025 and the European Association of Agricultural Economists 2025 conferences, IP-WSD 2025 workshop at Columbia University, 10th DENeB workshop at the Potsdam Institute for Climate Impact Research (PIK), and seminar participants at Imperial College London for their helpful comments. This work was supported by funds from the UK Economic and Social Research Council through the London Interdisciplinary Social Science Doctoral Training Programme [grant number ES/P000703/1].

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1 Introduction

Undernutrition imposes substantial long-run welfare costs on cognitive development and human capital accumulation (Glewwe et al., 2001; Maluccio et al., 2009), long-term health (Hoynes et al., 2016), and labour productivity and earnings (Case and Paxson, 2008). While progress has been made, Sub-Saharan Africa remains characterised by highly vulnerable malnourished populations, with 35.0% of children below five years old being stunted (i.e., height-for-age below two standard deviations from international standards; global average: 23.1%) (Takele et al., 2022). There are concerns that climate change and the increased frequency and intensity of extreme heat events may slow or even reverse progress against malnutrition. A growing body of work points towards reduced agricultural yields as one leading mechanism of the negative impact of extreme heat on nutrition, particularly in developing countries with a high prevalence of subsistence and smallholder farmers (Hoddinott and Kinsey, 2001; Block et al., 2022; Amondo et al., 2023; Stainier et al., 2025). Reduced household income and food price volatility have also been hypothesised as channels through which climate shocks translate into nutritional harm, but have not been established as causal pathways (Headey and Venkat, 2024; Blom et al., 2022).

Much of the climate-nutrition literature relies on cross-sectional health and household budget survey data, limiting the identification of causal relationships. Commonly used outcomes, such as self-reported food security and dietary diversity indices, are coarse. They capture extensive margins but not nutrient adequacy, quantities, or substitution patterns, obscuring the mechanisms that link climate shocks to diets and nutrition. Evidence on heterogeneity is also thin, with little assessment of which populations are most vulnerable or resilient. This ambiguity hampers policy design. Without credible causal pathways, it is unclear whether to prioritise productivity-enhancing agricultural adaptations, social protection to delink income from shocks, price stabilisation and insurance, or targeted nutrition programs.

This study compiles panel data from four Sub-Saharan countries, with diverse climates and a total of 19 waves, to (1) quantify the causal impact of extreme heat on food consumption and energy and nutrient availability; (2) investigate underlying pathways through households' agricultural production, income, and food prices; and (3) examine vulnerabilities, resilience, and adaptation behaviour.

Data from the World Bank Living Standards Measurement Study - Integrated Surveys on Agriculture (LSMS-ISA) in Malawi, Nigeria, Tanzania, and Uganda, collected between 2008 and 2023, are linked with high-resolution gridded hourly weather data. I harmonise these surveys to create a unique household panel dataset, from farm to fork, including information on agricultural production, livestock ownership, food consumption, food prices, child anthropometrics, and socio-economic and demographic information. The main econometric specification exploits the randomness of weather variations during the growing season after controlling for seasonality, location, macro-level shocks, and unobserved household characteristics. It leverages the fact that extreme heat reduces crop yield and that these losses could harm household welfare post-harvest. The dependent variables measure the availability and adequacy of diets. In order to more comprehensively assess diet quality, we consider calories along with the following set of nutrients: protein, iron, zinc, vitamin A, and folate. The treatment variables are a series of temperature bins measuring the share of days in the growing season with average temperatures belonging to specific 4°C-wide bins ranging from < 20°C to ≥ 32°C. Growing seasons are country and region-specific. I focus on how the last growing season temperatures impact consumption post-harvest but before the subsequent growing period starts, thus I abstract from heat's contemporaneous impacts on health or labour productivity. The unit of observation is panel households. To address potential confounders, I include household, region-year, and interview month-of-year fixed effects. Identification comes from within-household, season-to-season variation in unexpected temperature fluctuations during the growing season after netting out seasonality and macro shocks. I also control for total growing-season precipitations that can materially affect crop yields and food consumption. I show that results are robust to including alternative time trends (e.g., region-level) and non-growing season weather.

This analysis reveals that a one percentage point increase in the share of growing season days with an average temperature above 32°C decreases the daily availability of energy per adult equivalent unit by 1.32% relative to a one percentage point increase in the share of growing season days below 20°C. The impact on the availability of protein, zinc, and iron are of similar magnitude. It also reduces the share of households with adequate weekly availability of energy by 0.7 percentage points. On average, sample households experienced 2.2% of growing season days with an average temperature above 32°C and only 36% meet the adequacy threshold for weekly energy requirements. These results are particularly driven by significant reductions in the availability of energy from cereals, pulses, and vegetables, which form the bulk of diets.

Second, I show that a one percentage point increase in the share of growing season days with an average temperature above 32°C decreases total yields by 4.27%. I find that, in most cases, heat during the growing season causes a decline in home-grown consumption, particularly for pulses and vegetable proteins. I show that these proteins are among the key nutritional drivers of child stunting in this setting. Findings show limited evidence of adaptation by purchasing more food. Food purchases and nutrient availability from purchased foods actually decrease for most food groups, including non- or minimally-produced foods by households in the sample, like fish and seafood.

Third, the main underlying pathway through which temperature-related yield impacts translate to undernutrition is likely income. I find evidence that extreme heat during the growing season subsequently decreases total expenditure, a proxy for household income in developing settings. This effect is stronger for the poorest, who shift some non-food expenditure toward food. Hot temperatures during the growing season also reduce the share of crop production sold by households and increase family labour on the farm, crowding out time for outside income-generating options. Finally, I show that extreme heat increases the likelihood of livestock loss and decreases livestock sales, particularly for large ruminants, demonstrating that livestock holding does not play a buffering role for nutrition and income.

This paper makes several contributions. First, I add to a growing literature on heat's impact on welfare in developing contexts. While heat's impacts on yields are well documented (Schlenker and Roberts, 2009; Hultgren et al., 2025), crop losses are difficult to translate into societal welfare losses without understanding how households respond. I consider the impacts of weather shocks on home-grown and purchased foods separately, something that is largely absent from the literature (Amondo et al., 2023; Andrew Dillon and Oseni, 2015).

Second, I further the works of Bentze and Wollburg (2024) and McCullough et al. (2024) in compiling and integrating LSMS-ISA panel data across countries in Sub-Saharan Africa by additionally harmonising anthropometrics, livestock management, and community food prices information. This provides a public good for researchers and practitioners. Unlike previous works based on cross-sectional health survey data (Carpena, 2019; Stainier et al., 2025), this study relies on temporal weather variations within households to estimate the causal impact of extreme heat on household nutrition. The depth of the LSMS-ISA data also allows the investigation of heterogeneous patterns in vulnerabilities, as well as breaking down previously overlooked underlying pathways.

This is a work in progress. While preliminary results highlight the potential mitigating role of School Feeding Programs, I plan to further investigate the role of social protection schemes. I will also examine if increased experience of extreme heat mitigates the effect through long-run adaptation. Finally, I propose to disentangles demand-driven and supply-driven heat-induced price impacts, extending previous analyses in Sub-Saharan Africa (Emediegwu, 2024), and making a methodological contribution to measure climate impacts on commodity prices.

This paper is divided as follows. Section 2 details the data sources and provides summary statistics. Section 3 describes the main empirical strategy. Section 4 presents the main results. Section 5 investigates underlying mechanisms. Section 6 discusses the results and limitations and concludes.

2 Data

The analysis is based on the LSMS-ISA data, covering four Sub-Saharan African countries (Malawi, Nigeria, Tanzania, and Uganda) collected over 19 waves between 2008 and 2023.[1] I restrict the sample to panel households observed for at least three waves, with available geolocation, non-missing interview year and month, and at least two waves with non-missing and

[1]The LSMS-ISA panel data is also available from Ethiopia, Mali, and Niger, but is not included in this analysis. In Ethiopia, the food item list is quite limited in the first two waves, and the first wave excludes urban households. In Mali and Niger, only two waves of data are available, which limits panel method inference. Also, child anthropometric measurements are not available for Mali and Niger.

positive food consumption data.[2]

Bentze and Wollburg (2024) have harmonised the survey waves across countries into a unique dataset. It covers both rural and urban households and is nationally representative of the four countries in the sample. It includes household-level information on demographic structure, asset ownership, economic activities, and other socioeconomic characteristics. The harmonised dataset also contains plot- and crop-level agricultural information for each farming household.[3] I further harmonise household livestock ownership data, housing characteristics, and information on participation in social assistance programmes, individual-level health and labour participation characteristics, as well as anthropometric measurements (height and weight) for children aged 0 to 5 years old.

The surveys also collected household-level food consumption data for the seven days before the interview date. This includes a disaggregation by food item and the item source, i.e., own production, purchased, or received. It is important to note that the survey does not capture the exact quantity of food actually consumed, since some may be lost during cooking or wasted, nor does it reveal which household members ate specific foods. Thus, the measures reflect household-level food availability for consumption rather than actual individual intake. I match food items to their nutritional content using food composition tables and estimate household-level total energy and nutrient availability and requirement based on total consumption and household composition, following the work of McCullough et al. (2024).[4] I also construct adequacy percentages by dividing each household's total energy and nutrient availability by its recommended intake. Based on these percentages, I create dummy variables that indicate whether household-level caloric or nutrient consumption falls below 100, 80, or 60 percent of the recommended levels.

Figure 1 maps the LSMS-ISA cluster locations. Within-country geographical coverage is exhaustive. Spanning approximately 31° of latitude and crossing the equator, the clusters encompass diverse climates. Uganda has large highland regions with cooler temperatures. Nigeria

[2]See Appendix A for more detail on the data cleaning procedures.

[3]See Bentze and Wollburg (2024) for more detail on the harmonised LSMS-ISA dataset.

[4]For example, children and elderly members are counted as less than one full adult equivalent. See Appendix A for more detail.

and Tanzania have distinct wet and dry seasons, while Uganda and Malawi receive more consistent rain due to proximity to the equator. Finally, the north of Nigeria is hot and arid.

The dataset includes 43,151 observations from 11,664 unique households across 4,193 clusters (i.e., village, group of villages, or urban block). Nigeria provides the most observations, while Uganda includes the highest number of waves. The mean prevalence of household energy adequacy is 36%. It is the highest for Malawi and the lowest for Nigeria. Adequacy is the highest for Vitamin A and the lowest for folate.

Households' geocoordinates are matched with ECMWF ERA5's $0.1° \times 0.1°$ (equivalent to approximately 11km $\times$ 11km at the equator) gridded hourly temperature data (Hersbach et al., 2020). Precipitation data are constructed from the University of California, Santa Barbara Climate Hazards Center Infrared Precipitation with Station (CHIRPS)'s $0.05° \times 0.05°$ gridded monthly data (Funk et al., 2015), reaggregated using means to match ERA5's resolution and grids. The weather data for a specific grid square is assigned to an LSMS-ISA cluster if the cluster lies within that grid square.

Daily temperature data are categorised into temperature bins. These bins represent the number of growing season days that fall < 20°C, $[20, 24)$°C, $[24, 28)$°C, $[28, 32)$°C, and ≥ 32°C.[5] The binning approach enables flexibility in the analysis of the distribution of temperature exposure and has become the norm in the weather-health and climate economics literature (Dell et al., 2014; Deschenes, 2014). Country- and region-specific growing season dates are defined according to data from the Famine Early Warning System (FEWS) network.[6],[7] Given the variation in growing season length between settings, I harmonise the count variables into relative measures representing the share of days in each temperature range during the respective growing season. Table 1 shows that, on average across the four countries, households experience 29.9% of growing season days between 20 and 24°C, 50.4% between 24 and 28°C, 12.0% between 28 and 32°C, and 2.2% above 32°C. There is also significant across-country variation with hotter

[5]While some of the literature has used higher temperature thresholds to investigate the impact of heat exposure on malnutrition (e.g., Blom et al. (2022) use > 35°C as their highest temperature threshold in West Africa), it would be too high for the countries in this analysis. In addition, Banerjee and Maharaj (2020) and Deschênes and Greenstone (2011) have highlighted significant mortality above 32°C.

[6]Source: Famine Early Warning Systems Network, Season Calendar (Accessed 10 April 2025).

[7]Uganda has a bimodal agricultural season, only the first and main growing season is accounted for in this analysis.

temperatures in Nigeria and cooler in Tanzania.

3 Main empirical strategy

3.1 Food and nutrient consumption

The main specification consists of household-level multi-way panel fixed effects (FE) regressions.

$$y_{h,g,c,y,m} = \sum_b \beta_b T^b_{g,y} + \alpha_1 P_{g,y} + \alpha_2 P^2_{g,y} + \gamma_{c,y} + \lambda_m + \eta_h + \varepsilon_{h,g,c,y,m} \quad (1)$$

The dependent variables $y_{h,g,c,y,m}$ are nutritional outcomes for household h in geolocation g (i.e., cluster) in country c surveyed during month m following growing season y. These include energy and nutrient availability per adult equivalent and household energy and nutrient adequacy indicators at 100%, 80%, and 60% of the recommended levels. The main independent variables consist of the temperature bins $T^b_{g,y}$ that represent the share of the total number of days within the growing season in cluster g and growing season y that fall < 20°C (base), $[20, 24)$°C, $[24, 28)$°C, $[28, 32)$°C, and ≥ 32°C. The specification controls for total precipitation $P_{g,y}$ during the growing season and its square.

The analysis exploits the randomness of within-household temporal weather variations after controlling for seasonality, location, macro-level shocks, and unobserved household characteristics. Country-growing season FE ($\gamma_{c,y}$) account for any country-wide changes in temperature or nutrition that may be spuriously correlated over time. They also account for macro-level trends and potential changes in measurement error between LSMS-ISA waves. Interview month FE (λ_m) improve precision by controlling for seasonality in nutrition, particularly the difference in the timing of interview since harvest. Household FE (η_h) control for unobserved household characteristics and heterogeneity in temperature exposure and nutrition. For example, the Northern regions of Nigeria are both warmer and have higher levels of undernourishment.

The models are estimated using the computationally efficient estimator for linear regression using high-dimensional fixed effects developed by Correia (2016).[8] Estimates are weighted using household sampling weights. These reflect the inverse probability of selection into the sample and are adjusted to account for non-response and dropped observations.[9] I multiply the log energy and nutrient availability per adult equivalent unit outcome variables by 100 so the coefficients can be interpreted as percentage change. The coefficients for the household energy and nutrient adequacy specifications are to be interpreted as percentage point change. The standard errors are clustered at the household level to account for any correlation between the error terms of observations in the same household. For example, a household relying heavily on beans as its main protein source may repeatedly face similar shortfalls in protein intake across survey rounds when bean yields are reduced by heat, creating correlation in their error terms that household-level clustering accounts for.

3.2 Undernutrition outcomes

After estimating the effects on energy and nutrient availability, I want to quantify the potential impact on nutritional outcomes. For this, I focus on children below five years old, as it represents the age range in which the human body develops the most in terms of physical and cognitive features (Black et al., 2013). I follow a child-level dynamic Mundlak correlated random effects approach.

$$y_{i,h,t} = \alpha + \beta_0 y_{i,h,0} + \beta_1 y_{i,h,t-1} + \gamma F_{h,t-1} + \delta X_{i,h,t} + CRE_{i,h} + \varepsilon_{i,h,t} \tag{2}$$

The dependent variable $y_{i,h,t}$ is the child's height-for-age Z-score (HAZ), based on the 2006 World Health Organization (WHO) Child Growth Standards (World Health Organization, 2006). Stunting is defined as HAZ<2SD. It measures chronic undernutrition. Shocks reducing the HAZ of children are rarely offset by subsequent more rapid linear growth and are linked to poorer educational outcomes, congnitive skills, and lower future incomes (Black et al., 2013; Maluccio et al., 2009). Figure B3 shows the distribution of HAZ across the sample. The mean

[8] This estimator iteratively eliminates singleton groups, i.e., groups as defined by fixed effect levels with only one observation (Correia, 2016).

[9] See Appendix A for more detail.

prevalence of stunting is 32.3%, within the range of previous findings in Sub-Saharan Africa and aligned with the latest WHO estimates (Takele et al., 2022).[10]

The key explanatory variable of interest is $F_{h,t-1}$, which captures household food consumption in the previous period, measured in terms of household daily energy and nutrient availability. The lagged specification reflects that changes in dietary intake take time to materialize into child growth outcomes. The model also includes two terms that capture the persistence of child growth outcomes. $y_{i,h,0}$ represents the child's initial health endowment at the beginning of the observation period. Controlling for this initial condition helps address potential bias from unobserved early-life factors (e.g. birth weight, maternal health, prenatal nutrition) that influence both current nutritional status and subsequent growth trajectories (Jayachandran and Pande, 2017). The lagged dependent variable $y_{i,h,t-1}$ captures the dynamic nature of child growth, as HAZ is highly persistent over time. Including this term allows the model to account for state dependence, i.e. the fact that past nutritional status is a strong predictor of current status, and to better isolate the contribution of dietary intake and other covariates to changes in HAZ.

The vector $X_{i,h,t}$ includes individual- and household-level controls that may affect child growth, such as the child's age (in months) and sex, whether they are the eldest in the household, household head sex, age, and education, and a PCA-based household asset index. These variables account for observable heterogeneity in the determinants of child nutrition. The term $CRE_{i,h}$ represents the Mundlak correlated random effects, i.e. the household-level means of time-varying regressors. Including these averages allows the model to relax the strict exogeneity assumption of standard random effects and to control for unobserved time-invariant heterogeneity that may be correlated with the covariates. $\varepsilon_{i,h,t}$ denotes the idiosyncratic error term. Robust standard errors are clustered at the household level. The sample includes 4,541 unique children aged 6 to 59 months,[11] keeping only children with at least two waves with non-missing height measurements.[12]

Using the estimated relationships and the direct impact of extreme heat on energy and nutrient availability, I will be able to estimate the expected impact of future warming on child

[10] World Health Organization. Prevalence of stunting in children under 5 (%), 2024.
[11] Children below 6 months are excluded as they are likely to rely exclusively on breastfeeding.
[12] See Appendix A for more detail.

undernutrition through the food intake channel via back-of-the-envelope calculations.

4 Main results

4.1 Food and nutrient consumption

I estimate Equation 1 to test how heat impacts caloric and nutrient outcomes. Panel A of Figure 2 plots the coefficient estimates from the last growing season temperature bins, for the dependent variable log of daily energy and nutrient availability per adult equivalent unit. It shows that a one percentage point increase in the share of growing season days with an average temperature above 32°C decreases the daily availability of energy per adult equivalent unit by 1.32% relative to a one percentage point increase in the share of growing season days below 20°C.[13] It is important to note that a one percentage point increase in the share of growing season days with an average temperature above 32°C represents a 45% relative increase in such days, on average over the sample (Table 1). The effects are similar on protein, iron, zinc, and folate. However, the effect is not statistically significant on vitamin A.

Panel B of Figure 2 plots the coefficient estimates for the dependent variables representing dummies for household adequacy in meeting the weekly dietary requirements for energy and the five nutrients. Results are similar, as the likelihood of a household meeting the adequacy requirements at the various thresholds for calorie, protein, iron, zinc, and folate decreases with growing season temperatures. There are two notable differences. First, while a one percentage point increase in the share of growing season days with an average temperature above 32°C decreases the likelihood of 100% and 80% protein adequacy among sample households by 1.20% and 1.34%, respectively, the effect is smaller for the 60% threshold (-0.75%). Second, for folate, the negative effect is only statistically significant for the 80% and 60% thresholds.

The results are robust to the use of more granular FE, including region $\times$ year FE and region $\times$ interview month-of-year FE. They are also robust to the exclusion of Uganda (due to

[13] Table C1 shows the results for energy availability and adequacy in a table format.

its bimodal agricultural season) and the use of modified sample weights (Table C2).[14]

I further investigate the effect by food group. Following the UN Food and Agriculture Organization household dietary diversity score framework (FAO, 2011), I divide food consumption into 12 groups: cereals; roots and tubers; vegetables; fruits; meat, poultry, and offal; eggs; fish and seafood; pulses, legumes, and nuts; milk and milk products; oil/fats; sugar/honey; and a miscellaneous category. Figure B1 shows the composition of daily diets by country and food group. Cereals represent 46.4% of daily energy availability per adult equivalent unit in the sample, followed by oil and fats (17.0%), roots and tubers (13.0%), pulses, legumes, and nuts (11.3%), and meat, poultry, offal (2.3%). All the other food groups individually represent 2% or less of energy availability.

Panel A of Figure 3 presents the results for the extensive margin (i.e. household consuming or not the given food category during the week) and Panel B the results for the intensive margin (i.e. total energy availability per adult equivalent unit for the given food group). The probability of consuming meat, poultry, and offal and eggs decreases while the probability of consuming pulses, legumes, and nuts and sugar/honey increases with growing season temperatures. On the other hand, a one percentage point increase in the share of growing season days with an average temperature above 32°C decreases the daily energy availability per adult equivalent unit by 1.52% for cereals, 3.60% for vegetables, 1.88% for meat, poultry, and offal, 1.56% for fish and seafood, and 1.56% for pulses, legumes, and nuts, while it increases the daily energy availability per adult equivalent unit by 2.05% for milk and milk products.

4.2 Heterogeneity

I am interested in exploring heterogeneity in the effect of last growing season temperatures on diets to identify the most vulnerable populations. Figure 4 presents heterogeneity by wealth (as proxied by a PCA-based household asset index), head sex and education, and the presence of children below five years old in the household. Panel A presents the effects on energy availability per adult equivalent unit. Panel B presents the effects for protein. While non-statistically

[14] Results are also robust to the use of Conley (1999)'s spatial heteroscedasticity and autocorrelation consistent (HAC) standard errors with a Barlett spatial weighting kernel with a distance cutoff of 100km and time lag of three years from the interview date. Results are available upon request.

different, the energy and protein availability of wealthier households and households with at least one child below five years old appear less impacted by extreme heat in the growing season. On the other hand, households with female heads appear more impacted (not a statistically significant difference).

4.3 Undernutrition outcomes

Now that I have established the impact of temperatures in the last growing season on energy and nutrient availability, I examine the importance of these dietary components in children's chronic malnutrition status, as measured by HAZ. As expected, more energy or nutrient availability is associated with higher HAZ and a lower likelihood of child stunting. In line with the irreversibility of shocks to child heights (Black et al., 2013), Table C3 confirms the expected positive sign on the baseline outcome variable, i.e. a child previously stunted has a high probability to remain stunted. As I control for this initial endowment, the lagged outcome variable reflects an error-correction (mean-reversion) mechanism. It picks up short-run deviations around the initial level. Children who are temporarily above or below their trajectory at $t-1$ tend to move back toward it at t. Dropping the initial endowment turns the coefficient on the lagged outcome variable from negative to positive as it then absorbs the persistent component.[15]

Figure 5 shows that protein, zinc, and folate are negatively associated with child stunting. Particularly, a child in a household above the 100% adequacy requirements for weekly protein availability has a 5.7 percentage point lower likelihood of being stunted in the following period. This is in line with the public health literature. Protein is essential for tissue growth. Zinc is crucial for cell growth and immune function. Finally, folate is critical for cell division and neurological development, especially in this setting where diets are low in animal-sourced foods (Black et al., 2013).

Prolonged heat exposure could lead to increased diarrheal disease and malaria. The former could be more pronounced in households with poor sanitation, leading to more hospitable environments for disease-carrying pathogens. The results are robust to adding controls for child health - symptoms of diarrhea and fever (a common symptom of malaria) in the last seven

[15] Results available from the author upon request.

days - and household sanitation - whether households have access to piped water and improved latrines.[16]

5 Potential mechanisms

This section aims to understand the pathways through which temperature-related crop losses affect household nutrition. I explore the role of domestic agricultural production on food consumption. I investigate the impact of extreme heat on household income and total expenditure including through shift from non-food expenditure to food expenditure, reduced sale of crop production, labour supply on the farm, and the role of livestock holdings as nutrient-rich food sources and income buffers. Finally, I aim to examine the impact of temperature-related yield reductions on food prices.

5.1 Agricultural yields

I first test whether hot temperatures impact crop yields. The study relies on Bentze and Wollburg (2024)'s harmonised LSMS-ISA information on household harvested quantities per plot. Yield is built from self-reported total harvested quantities divided by the total planted plot area during the last main agricultural season. The model specification follows Equation 1 with the logarithm of total yield and total harvest quantities as dependent variables. I multiply the log dependent variables by 100 so the coefficients can be interpreted as percentage change.

Figure 6 shows the effects of hot temperatures during the growing season on yields and harvested quantities. I find statistically significant decreases in overall yields for days above 24°C.[17] A one percentage point increase in the share of growing season days with an average temperature above 32°C decreases total yields by 4.27% and total harvested quantities by 3.47% relative to a one percentage point increase in the share of growing season days below 20°C. This

[16] Results available from the author upon request.

[17] While this threshold may appear low compared to other studies in the literature (Schlenker and Roberts, 2009; Hultgren et al., 2025), here, I study the effect of average temperatures, not maximum temperatures.

is 4.2 times larger than the effects of an increase in the number of days between 24 and 28°C. These results are consistent with past literature (Block et al., 2022).[18]

Now that I have established that hot growing season temperatures negatively impact crop yields, I test whether the observed negative impact on energy and nutrient availability is driven by reduced consumption from own-production. Figure 7 presents the effects of last growing season temperatures on energy and nutrient availability by source of consumption, between own-production and purchased food. Reductions in nutrient availability are equally driven by decreases in own-produced and purchased consumption, except for protein, for which reductions appear mostly driven by own-production. On the other hand, the negative effect of extreme heat during the growing season on energy availability is only statistically significant for purchased consumption. Figure B4 presents the results for energy availability per adult equivalent unit by food group and source of consumption.[19] Results are aligned with Figure 7, showing that the drop in energy availability from cereals (representing more than half of energy availability) is driven by reduced purchases. The same pattern is found for vegetables, meat, poultry, and offal, fish and seafood, and pulses, legumes, and nuts. Only the purchased energy availability from fruits increases with extreme heat during the growing season.

5.2 Income

The results from Figure 7 show that households are not compensating for reductions in energy and nutrient availability from own-produced food sources by purchasing more food. They even reduce the consumption of purchased food for most food groups, including some which are not or minimally produced by households in the sample, like fish and seafood. This is in line with an income shock channel, where extreme heat in the growing season may reduce income from the sale of agricultural output or the labour income from salaried work in agriculture or other impacted sectors.

[18] Figure B5 show the results disaggregated by country.

[19] Figure B2 shows the composition of daily energy availability per capita for each food group by source of consumption. More than 55% of calories consumed for each food group are purchased. Notable food groups with high share of own-produced energy availability are roots and tubers, cereals, fruits, and pulses, legumes, and nuts. A minor portion of calories is consumed from food received for free, except for fruits for which it represents approximately 18% of calories consumed.

The LSMS-ISA data does not contain household income information. As others in developing settings, I use total expenditure as a reliable proxy for permanent income (Carletto et al., 2021). Figure B8 shows that Engel's Law is empirically verified in the sample, as the share of total expenditure devoted to food decreases with total expenditure. I find that a one percentage point increase in the share of growing season days with an average temperature above 32°C decreases total expenditure by 1.80%, food expenditure by 2.47%, and non-food expenditure by 1.61% relative to a one percentage point increase in the share of growing season days below 20°C (Table 2). The latter results could be explained by both a decrease in available income and a reassignment of disposable income from non-food to food expenditure to compensate for the decrease in own-produced food consumption sources or weather-induced increases in food prices (which will be assessed at a later stage).

Reduced sales of own-produced crops amplify the income losses from extreme heat during the growing season. Consistent with this mechanism, Table 2 shows that a one–percentage-point increase in the share of growing-season days with mean temperature above 32°C reduces the share of output sold by 2.3%. In parallel, farming households substitute toward family labour on the farm during hotter seasons and scale back hired labour; nevertheless, total on-farm hours increase by 3.3% for the same change in heat exposure (Table C4), crowding out time available for other income sources. Because less than half of farming households hire any labour - reflecting the predominance of smallholder farmers in the sample - the scope to buffer heat shocks by cutting hired work is limited.

5.3 Livestock

Livestock holdings can offer both a source of income through the sale of livestock products (e.g., milk, eggs, meat) and the sale of livestock. It also represents a valuable source of nutrient-rich food, particularly quality protein in undernourished settings. I examine whether extreme heat impacts households' livestock holdings, particularly the number of heads managed, purchased, sold, and lost, including by death. As livestock holdings data is collected over a 12-month recall period, I redefine my weather variables to match this timeframe. I also recentre the reference bin, and redefine temperature bins as the number of days in the last 12 months, given the standardised recall period. I estimate Equation 1 replacing the dependent variable with livestock holding variables and use a Poisson pseudo-maximum likelihood model to account for the high prevalence of zero-values.[20]

An additional day with a mean temperature above 34°C in the last 12 months reduces the number of large ruminant heads (e.g., cow, bull) under management by 4.67% and small ruminant heads by 6.33% (e.g., goat, sheep) relative to one more day in the reference bin 22-25°C. Findings show a negative impact of extreme heat in the last 12 months on the number of large ruminant heads sold and a positive impact on the number of large ruminant heads lost or dead. Contrary to other settings that have been studied, like Mongolia (Groppo and Kraehnert, 2016), cold temperatures are only rarely reaching below freezing in the sample countries, explaining the non-positive impact of temperatures below 10°C on heads lost (Figure 8).

Taken together, these findings indicate that, in the context of these four countries, livestock does not function as an effective buffer against heat shocks. Because animal-source foods account for less than 5% of household energy availability (Figure B1), the direct dietary-substitution channel is limited. At the same time, extreme heat reduces herd size through higher losses and fewer heads under management and is associated with fewer animals sold, shrinking the marketable surplus and dampening the income-smoothing role of livestock. In short, rather than offsetting the adverse effects of hot growing seasons, livestock holdings appear vulnerable to heat stress in both the asset and income pathways, thereby amplifying rather than mitigating potential nutrition risks.

5.4 Food prices

The LSMS-ISA survey collects food prices from the main market in each cluster in the month during which households are surveyed. I have additionally cleaned and harmonised these commodity prices per kilogram at the community level, including maize, sorghum, millet, onion,

[20]The presence of many structural zero values can lead to biased ordinary least squares estimates and inflated standard errors. Instead, Poisson pseudo-maximum likelihood models the conditional mean of the response variable. This approach is robust to distributional misspecification and is not restricted to count data. It is also robust to overdispersion as we use cluster-robust standard errors at the household-level. We specifically use the iteratively reweighted least-squares algorithm developed by Correia (2016) for Poisson regression models with multiple high-dimensional fixed effects. Singleton and separated observations are dropped.

banana, beef, goat meat, egg, dried and fresh fish, horsebean, groundnut, fresh milk, cooking oil, and sugar.

First, I plan to use a multi-way panel fixed effect approach, similar to Equation 1, to regress local temperatures during the last growing season on post-harvest local food prices. This will confound the demand and supply effects.

In another specification, prices in non-producing markets $p_{m,t}$ are regressed on local weather during the last growing season $\mathbf{W}^{\text{local}}_{m,t}$ in market m and time period t - capturing local demand impacts - and on a weather index during growing months among producing regions $\tilde{\mathbf{W}}^{\text{prod}}_{m,t}$ (weighted by inverse-distance to market m and long-run average production levels)[21] - capturing supply impacts. The model includes country $\times$ year ($\eta_{c,y}$) and market $\times$ month-of-year fixed effects ($\theta_{m,moy}$). Robust standard errors are clustered at the market level.

$$p_{m,c,t} = \alpha_1 + \beta_1^D \, \mathbf{W}^{\text{local}}_{m,t} + \beta_1^S \, \tilde{\mathbf{W}}^{\text{prod}}_{m,t} + \eta_{c,y} + \theta_{m,moy} + \varepsilon_{m,c,t} \tag{3}$$

This approach tests for market integration and disentangles demand-driven and supply-driven heat-induced price impacts, extending previous analyses in Sub-Saharan Africa (Emediegwu, 2024). This is a work-in-progress.

6 Discussion

This paper provides causal evidence that unusually hot growing seasons reduce household food availability and worsen undernutrition in Sub-Saharan Africa. I show that higher temperatures lower crop yields and depress daily energy availability, increasing the share of households with inadequate availability of calories and key nutrients, such as protein, iron, zinc, and folate. The impacts are largest for protein, a macronutrient I document as a salient determinant of child stunting in this setting. The evidence points primarily to an income pathway. Extreme heat

[21] Based on data from the International Food Policy Research Institute (IFPRI), Spatial Production Allocation Model (SPAM), 2024. (Accessed 3 April 2025).

reduces total household expenditure, lowers the share of agricultural output sold, shifts labour towards own-farm activities, and raises livestock losses. While wealthier households partially offset shortfalls in own-produced foods by purchasing more, resource-constrained households have considerably less scope for such adjustment.

A key contribution is to move beyond coarse categorical indicators such as dietary diversity scores and self-reported food security measures. By recovering nutrient-specific quantities, I reveal heterogeneity in the composition of losses that would otherwise be obscured. Equally important, all core variables, including yields, livestock holdings, food consumption, food prices, and nutritional outcomes, are measured within the same household-panel architecture, reducing scope for ecological fallacy and improving internal consistency of the estimated pathways.

The results complement and extend recent literature in Sub-Saharan Africa. Unlike studies based on self-declared climate shocks (e.g., Wollburg et al. (2024)), the identification strategy here leverages exogenous within-household weather variation and rich fixed effects, thereby improving causal interpretation. The prominence of protein losses in child stunting aligns with Khonje and Qaim (2024). The latter estimated the association between consuming or not consuming animal-sourced food and the likelihood of child stunting. I go further by quantifying intensive-margin responses and tracing direct links to specific nutrients rather than broad food groups alone. Moreover, the analysis incorporates dynamics, accounting for lagged nutritional status and the persistence of short-run intake shocks, thereby better aligning estimated short-run intake impacts with medium-run anthropometric risk.

Ongoing analysis of price mechanisms—both demand-driven (local purchasing pressure) and supply-driven (harvest shortfalls)—seeks to disentangle whether observed intake losses are amplified by local price spikes or primarily reflect income contractions. Early evidence of mitigation through school feeding suggests that targeted, protein-rich safety nets can be effective shock absorbers. Policy implications follow directly. On-farm adaptation should be paired with income-smoothing instruments that operate when adaptation margins are exhausted. Because protein losses are particularly pronounced, nutrition-sensitive interventions (e.g., school feeding, subsidies for protein-dense foods, fortified staples) are likely to yield positive returns in heat-affected seasons.

The study has limitations. Food consumption is measured over a 7-day recall window, which tends to overstate mean intake relative to longer recall periods (Mukherjee and Chaudhury, 2020). While this affects levels more than differences, it could bias estimated responses if recall behaviour covaries with heat. Because food may not be distributed equally within households (D'Souza and Tandon, 2019), the effects of extreme heat on undernutrition may be attenuated or masked for vulnerable subgroups, particularly if their allocation varies under financial stress.

Several avenues merit further inquiry. First, I aim to further explore heterogeneity and moderators. Second, I will study long-run adaptation via exposure experience, examining if households in historically hotter zones exhibit attenuated responses. Third, I will quantify the role of off-farm labour in stabilising incomes. Finally, I aim to investigate market integration and the price channel, decomposing supply-side harvest shocks from demand-side purchasing responses.

Future increased extreme heat frequency is expected to worsen chronic malnutrition in Sub-Saharan Africa, which may slow or reverse the progress made over the last decades. This finding, along with the identification of key mechanisms, participates in informing policymaking to design interventions to better support vulnerable populations under climate change, particularly in settings grappling with extreme poverty and malnourishment. Future research could hopefully make use of higher temporal variations with longer longitudinal panel data, including high-frequency individual-level dietary information, to further the understanding of linkages in the heat-malnutrition relationship in underdeveloped settings.

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Figures

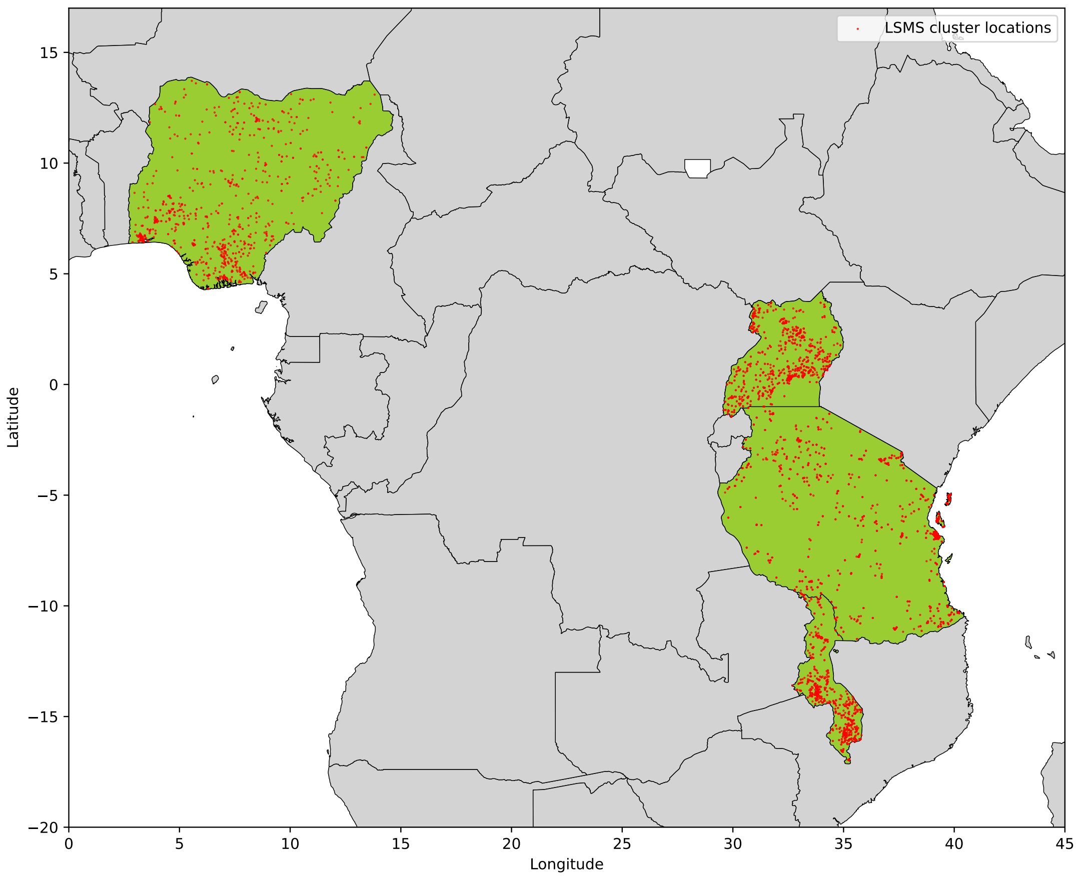

Figure 1. **Map of sample cluster locations.** Cluster locations are displayed in red. Based on 11,664 unique sample panel households in 4,193 unique clusters. The map of Africa is truncated to zoom on countries included in this analysis, i.e., Malawi, Nigeria, Tanzania, and Uganda (in green).
Back to Section 2.

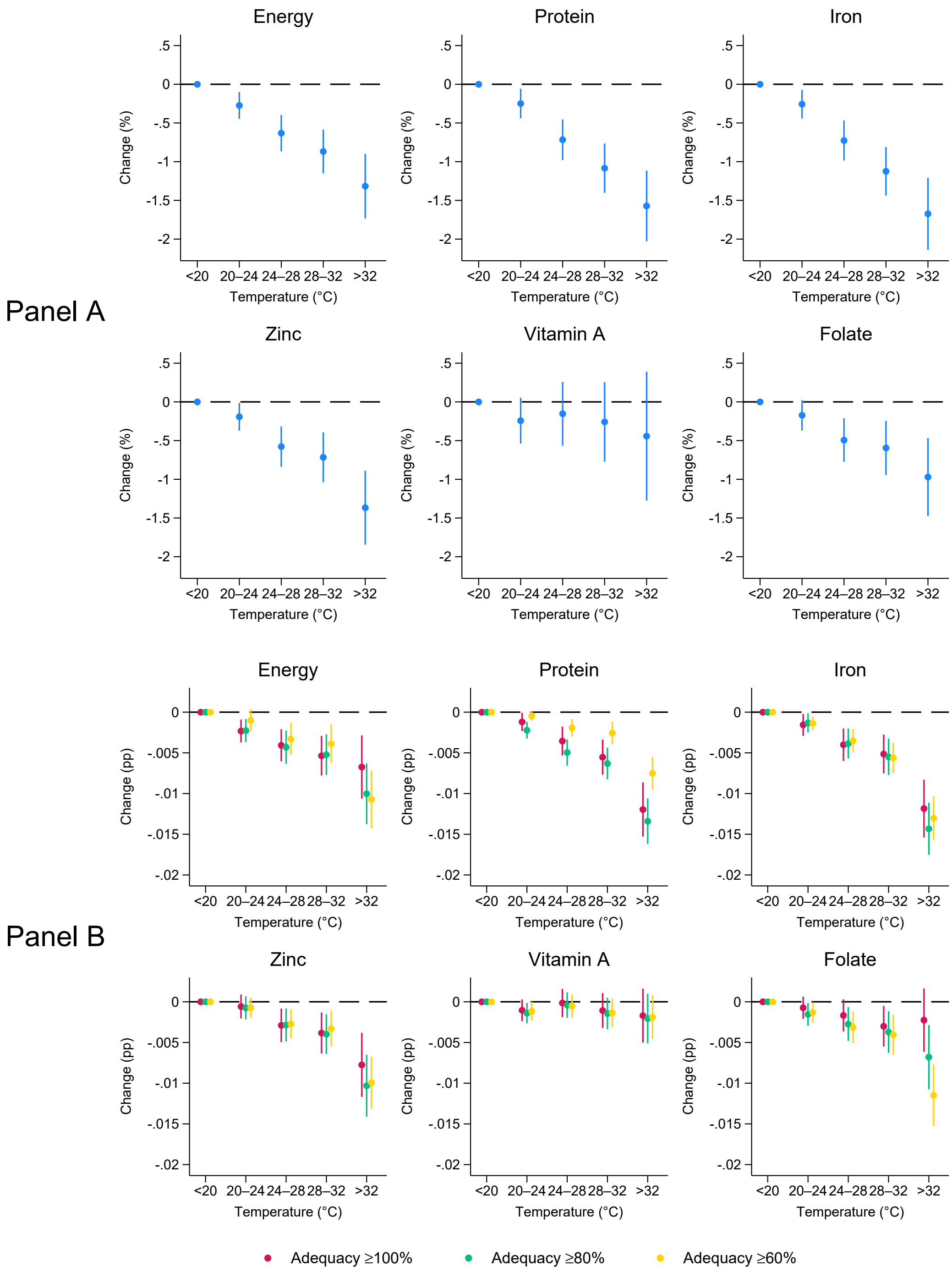

Figure 2. **Last growing season temperatures and energy and nutrient availability (Panel A) and adequacy (Panel B).** It represents the effect of a 1 percentage point increase in the share of days compared to the reference bin ($< 20°C$). Panel A coefficients are to be interpreted as percentage change and Panel B coefficients are to be interpreted as percentage point changes. Adequacy represents households above 100%, 80%, and 60% of weekly dietary requirements, estimated for each household based on age and sex composition. Confidence intervals for the 95% based on robust clustered standard errors at the household level. Sample weights are used. Back to Section 4.1.

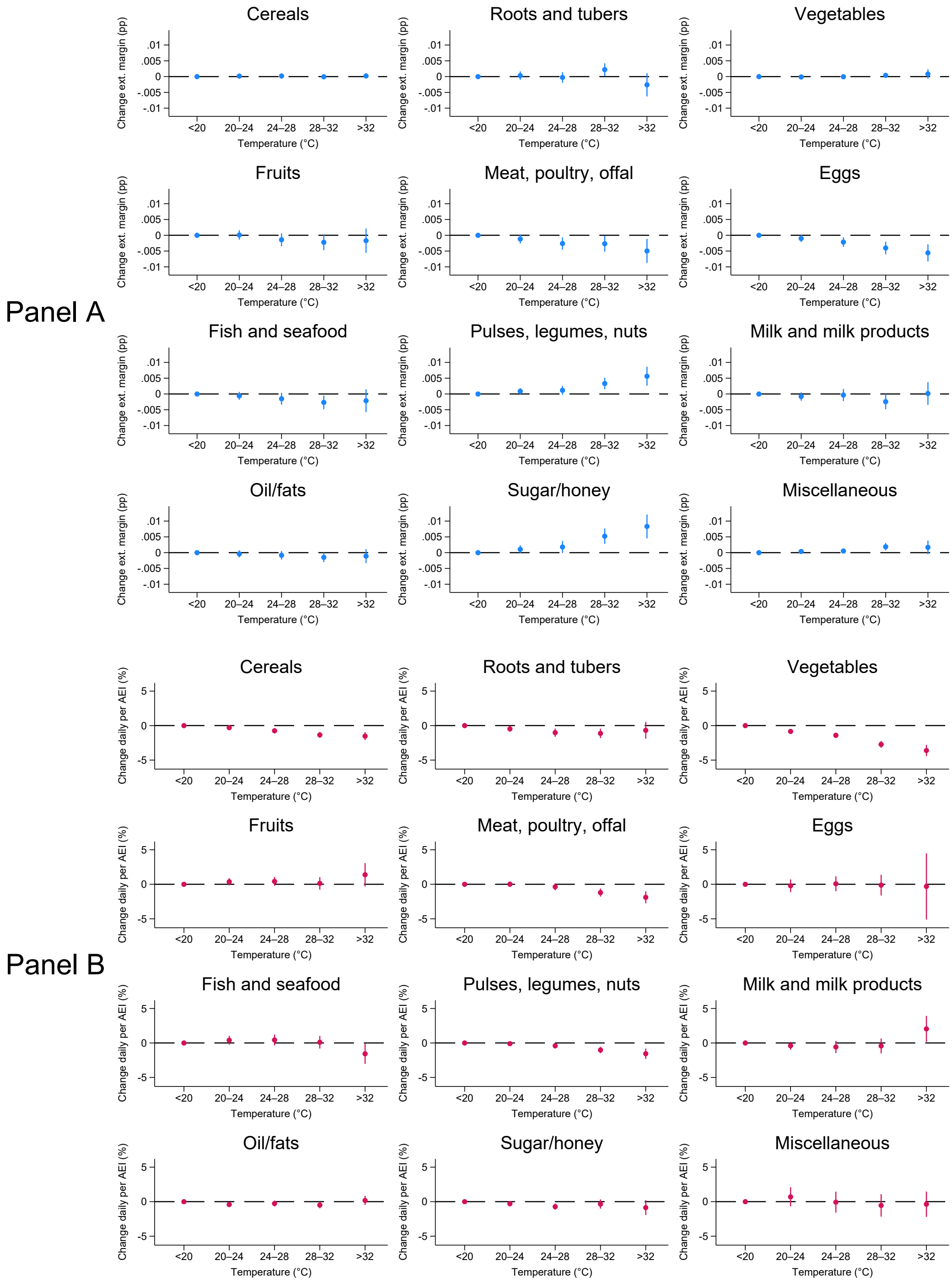

Figure 3. **Last growing season temperatures and consumption (Panel A, extensive margin) and energy availability (Panel B, intensive margin), by food group.** It represents the effect of a 1 percentage point increase in the share of days compared to the reference bin ($< 20°C$). Coefficients are to be interpreted as percentage change. Sample weights are used. AEI: adult equivalent index.

Back to Section 4.1.

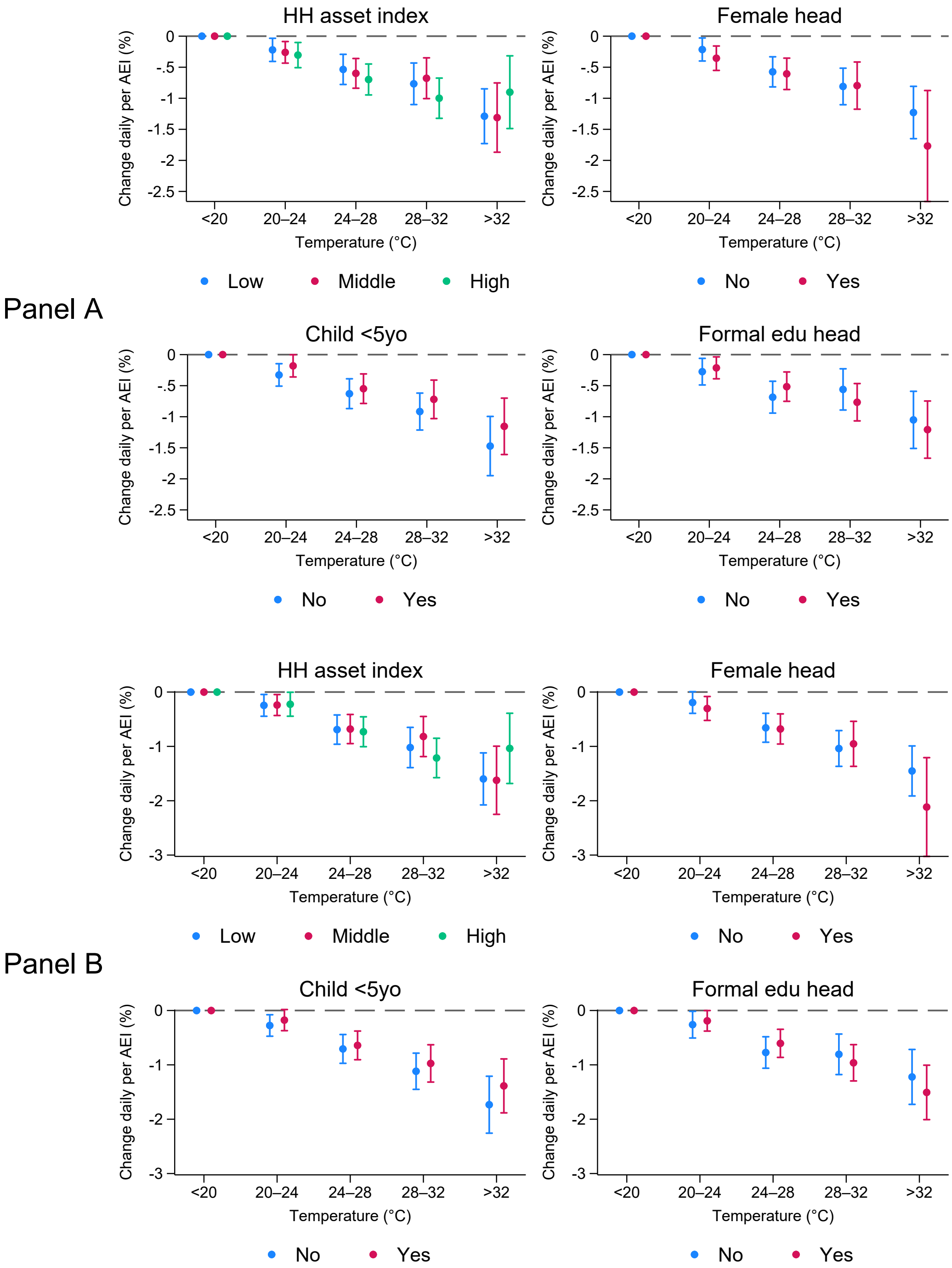

Figure 4. **Heterogeneous effects of last growing season temperatures on energy (Panel A) and protein availability (Panel B).** It represents the effect of a 1 percentage point increase in the share of days compared to the reference bin (< 20°C). Coefficients are to be interpreted as percentage change. Sample weights are used. AEI: adult equivalent index; HH: household; yo: years old.

Back to Section 4.2.

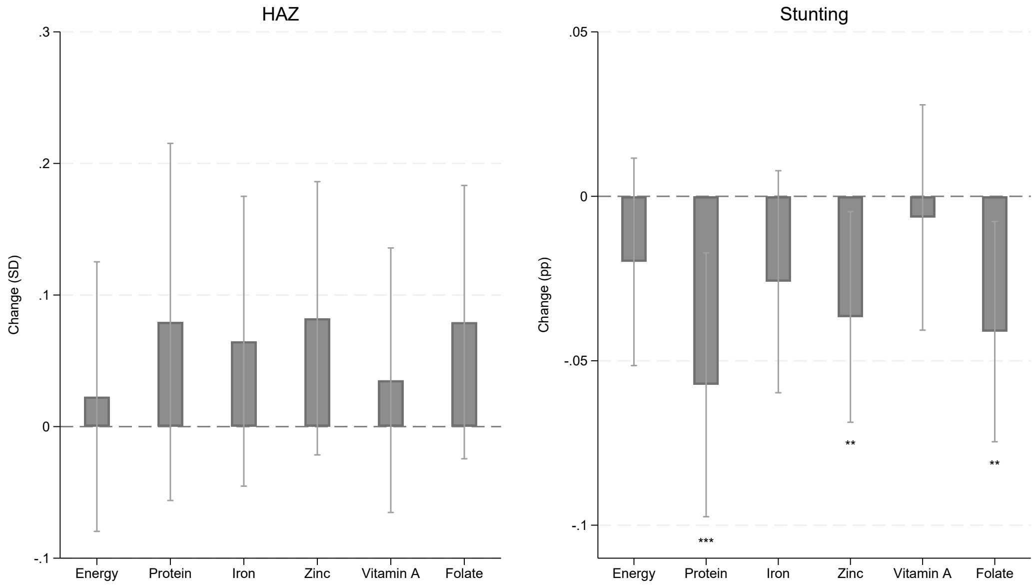

Figure 5. **Mundlak correlated random effects results on the impact of last period energy and nutrient adequacy on HAZ and stunting.** Plotting the coefficients on the lag 100% household adequacy variable.

Back to Section 4.3.

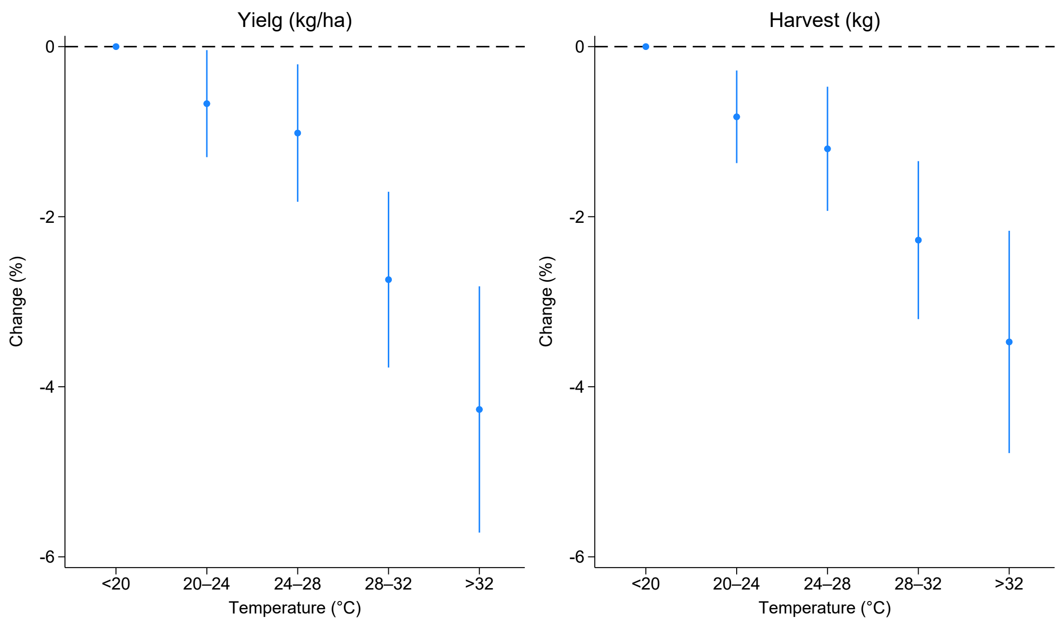

Figure 6. **Growing season temperatures on total yield and harvested quantities.** Coefficients are to be interpreted as percentage change. Sample weights are used. ha: hectare; kg: kilogram.

Back to Section 5.1.

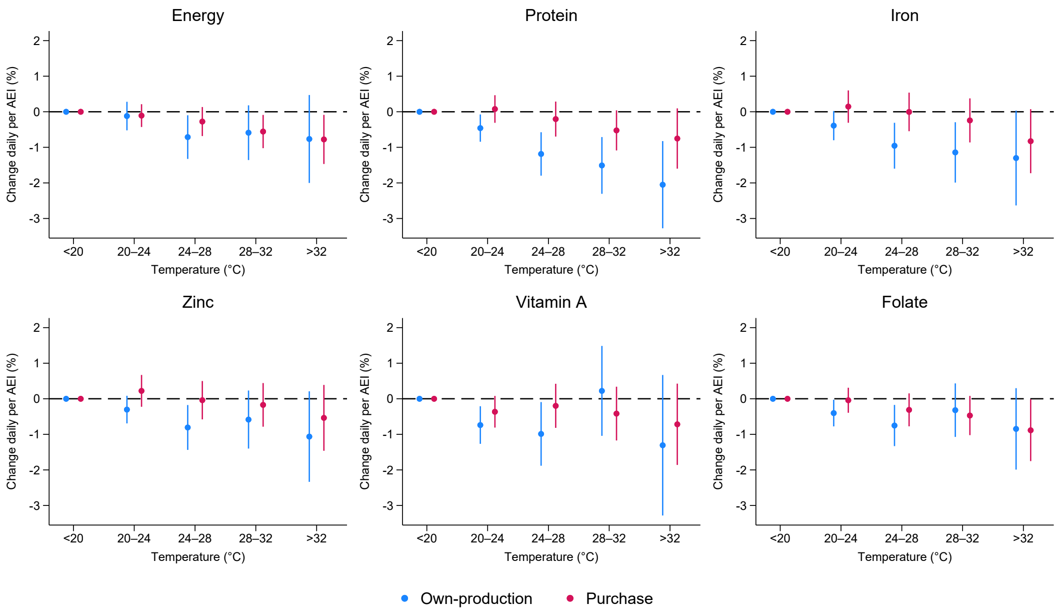

Figure 7. **Last growing season temperatures and energy and nutrient availability, by source of consumption.** It represents the effect of a 1 percentage point increase in the share of days compared to the reference bin ($< 20°C$). Coefficients are to be interpreted as percentage change. Sample weights are used. AEI: adult equivalent index.
Back to Section 5.1.

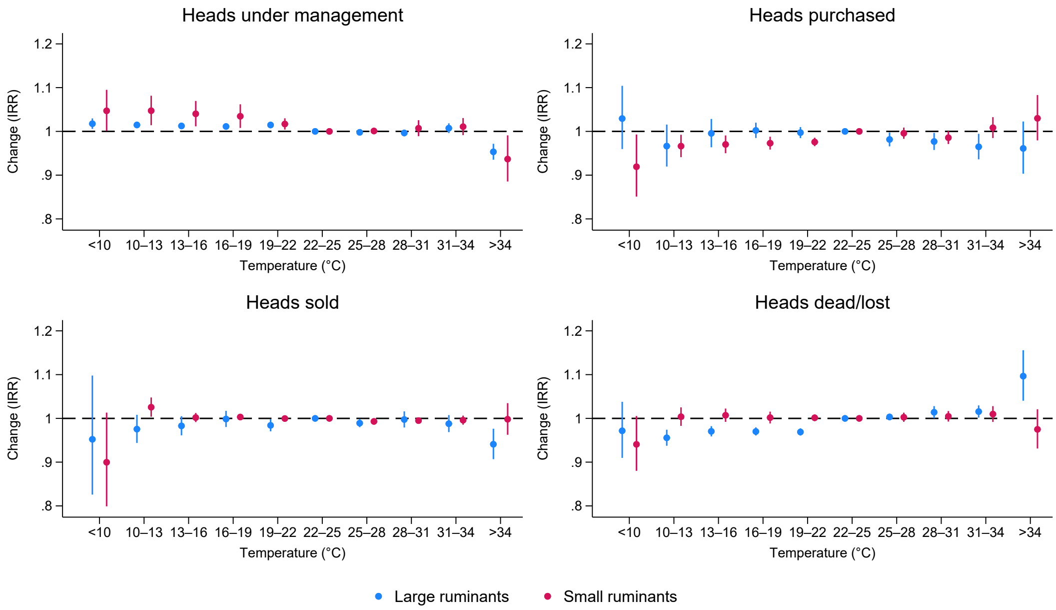

Figure 8. **Last 12-month temperatures and livestock under management.** PPML specification uses the computationally efficient estimator for Poisson regressions using high-dimensional fixed effects developed by Correia (2016). PPML coefficients are exponentiated and should be interpreted as IRR. Sample weights are used. IRR: Incidence risk ratio.
Back to Section 5.3.

Tables

	Malawi	Nigeria	Tanzania	Uganda	Total
Adequacy ($\geq 100\%$)					
Energy (%)	0.49	0.31	0.42	0.47	0.36
Protein (%)	0.86	0.68	0.88	0.83	0.74
Iron (%)	0.57	0.51	0.78	0.69	0.58
Zinc (%)	0.75	0.41	0.61	0.47	0.46
Vitamin A (%)	0.51	0.88	0.42	0.61	0.76
Folate (%)	0.64	0.32	0.29	0.64	0.40
Growing season weather					
$<$ 20C (%)	4.8	0.2	21.6	14.4	5.5
20-24C (%)	58.5	11.8	52.8	68.8	29.9
24-28C (%)	32.7	67.1	24.3	14.7	50.4
28-32C (%)	3.8	17.5	1.3	2.0	12.0
$>$ 32C (%)	0.2	3.4	0.0	0.0	2.2
Precipitations (mm)	188.2	155.2	84.0	149.8	148.3
Observations					
Number of waves	4	5	3	7	19
Number of clusters	926	811	1,396	1,060	4,193
Number of households	1,860	4,475	2,647	2,682	11,664
Number of observations	6,768	15,490	7,549	13,344	43,151

Table 1. **Summary statistics.** Sample weights are used.
Back to Section 2.

	Total exp	Food exp	Nonfood exp	Share prod sold
20–24C	-0.140	-0.264*	0.389**	1.000
	(0.097)	(0.146)	(0.196)	(0.004)
24–28C	-0.106	-0.534***	0.582**	0.996
	(0.142)	(0.194)	(0.259)	(0.005)
28–32C	-0.891***	-1.286***	-0.181	0.993
	(0.191)	(0.238)	(0.306)	(0.007)
$>$ 32C	-1.797***	-2.474***	-1.613***	0.977**
	(0.283)	(0.334)	(0.409)	(0.011)
N	32887	21666	21654	12211
R^2	0.850	0.796	0.804	0.145
Household FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes	Yes
Precipitations	Yes	Yes	Yes	Yes
Model	OLS	OLS	OLS	PPML

Table 2. **Last growing season temperatures and expenditure and the sale of produced crops.** For expenditure regressions, coefficients are to be interpreted as percentage change. Malawi is excluded because data is only available for the first two waves. Disaggregated data on total expenditure between food and non-food items is missing for Nigeria wave 5 and for Uganda. Food expenditure includes a valuation of own-produced food consumption at market prices. All expenditure values are in 2020 USD. The share of sold production is only available for farming households and contains a significant portion of missing values. The PPML specification uses the computationally efficient estimator for Poisson regressions using high-dimensional fixed effects developed by Correia (2016). For PPML, R^2 is a pseudo-R^2 and coefficients are exponentiated and should be interpreted as IRR. It drops separated observations from the estimation sample to avoid problems of perfect fit in likelihood estimations. Sample weights are used. IRR: Incidence risk ratio; PPML: Poisson pseudo-maximum likelihood.
Back to Section 5.2.

Appendix

A. Appendix Data Details

Weather data

TBA.

LSMS-ISA panel data

TBA.

Nutrient data

TBA.

Agricultural output data

TBA.

Anthropometric data

TBA.

Price data

TBA.

B. Appendix Figures

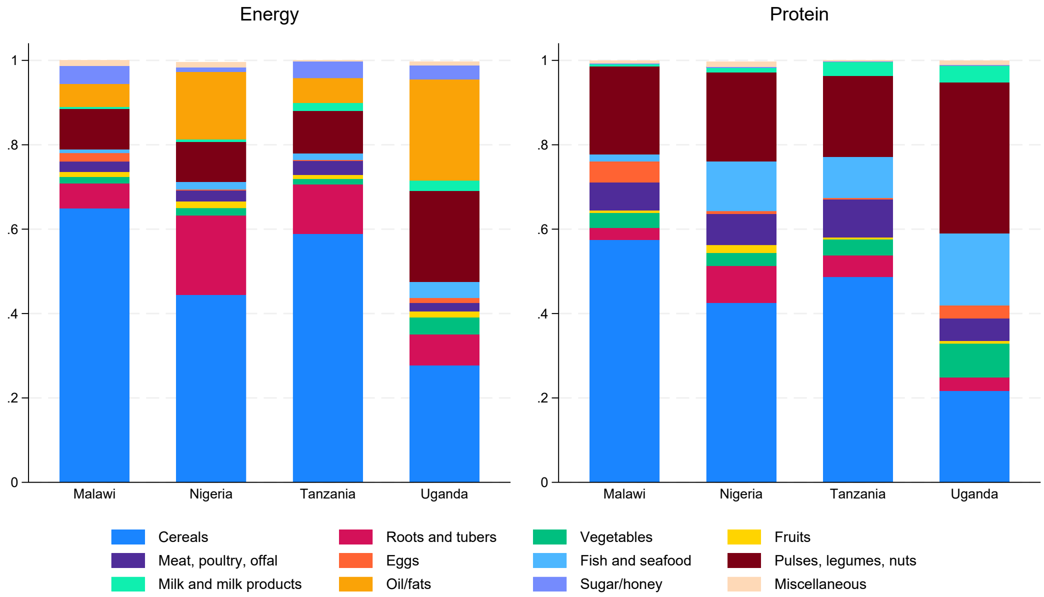

Figure B1. **Food group composition of daily energy and protein availability per adult equivalent unit, by country.** Sample weights are used.
Back to Section 4.1.

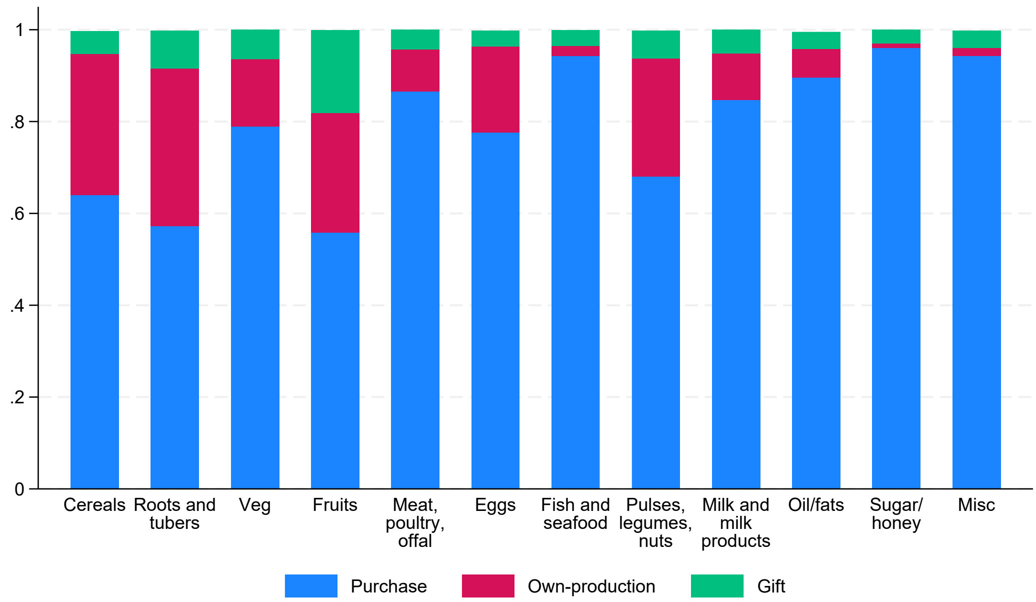

Figure B2. **Source of daily energy availability per adult equivalent unit, by food group.** Sample weights are used.
Back to Section 5.1.

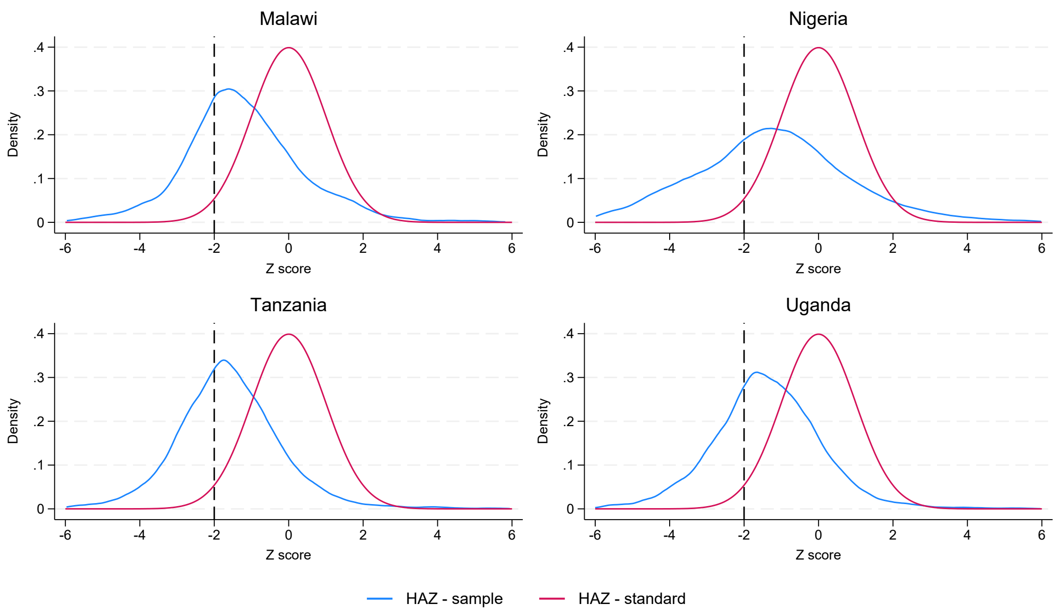

Figure B3. **Density plots of height-for-age Z-scores, by country.** Based on the full sample of panel households without missing information. Z-scores based on the 2006 World Health Organization Child Growth Standards (World Health Organization, 2006). Standard represents the standard normal distribution. The vertical dotted line shows the definition of child stunting, i.e., HAZ<2SD.
Back to Section 3.2.

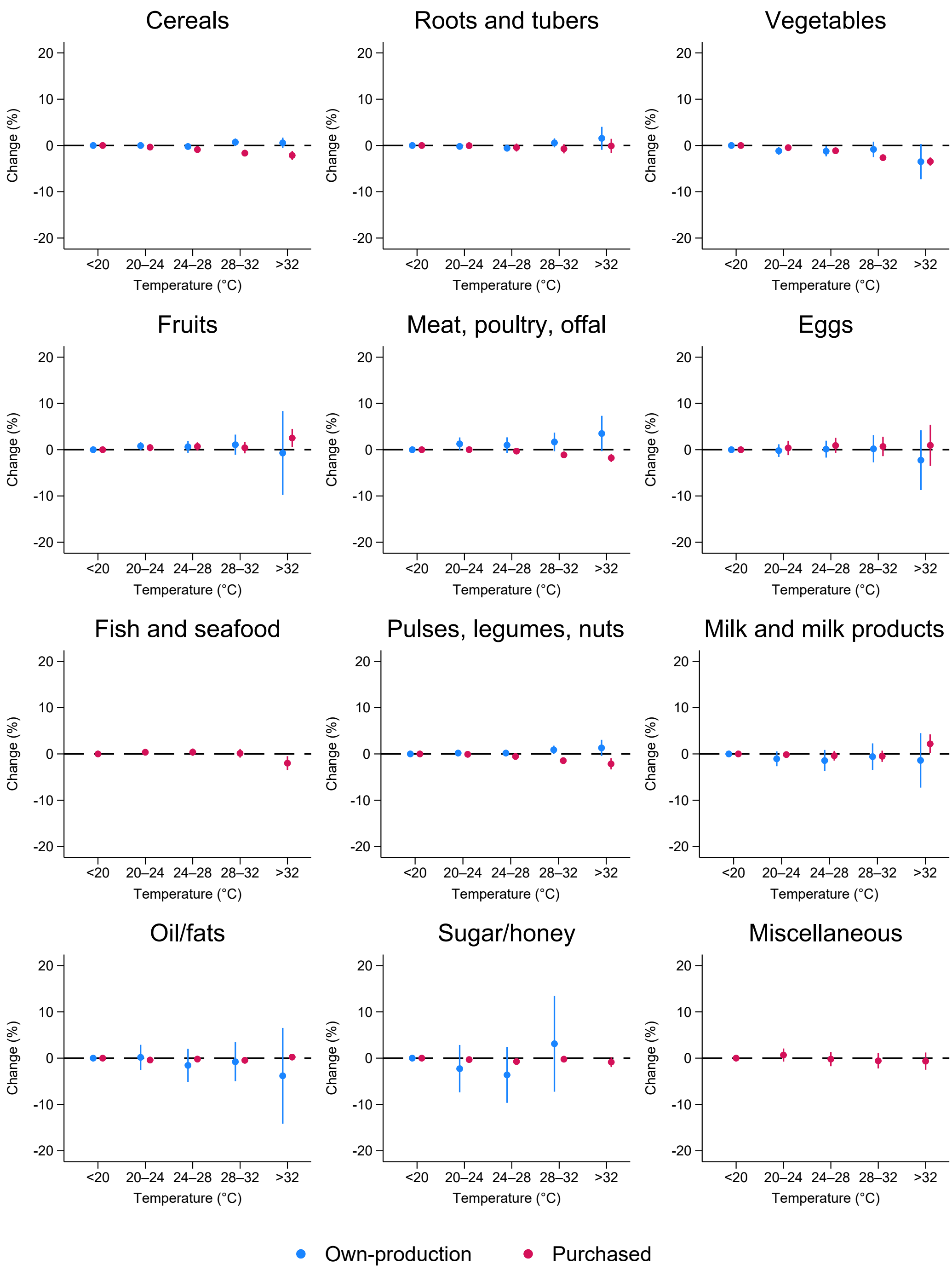

Figure B4. **Last growing season temperatures and energy availability for each food group, by source of consumption.** I omit own-production for fish and seafood and miscellaneous due to low household-level production in the sample. It represents the effect of a 1 percentage point increase in the share of days compared to the reference bin ($< 20°C$). Coefficients are to be interpreted as percentage change. AEI: adult equivalent index.
Back to Section 5.2.

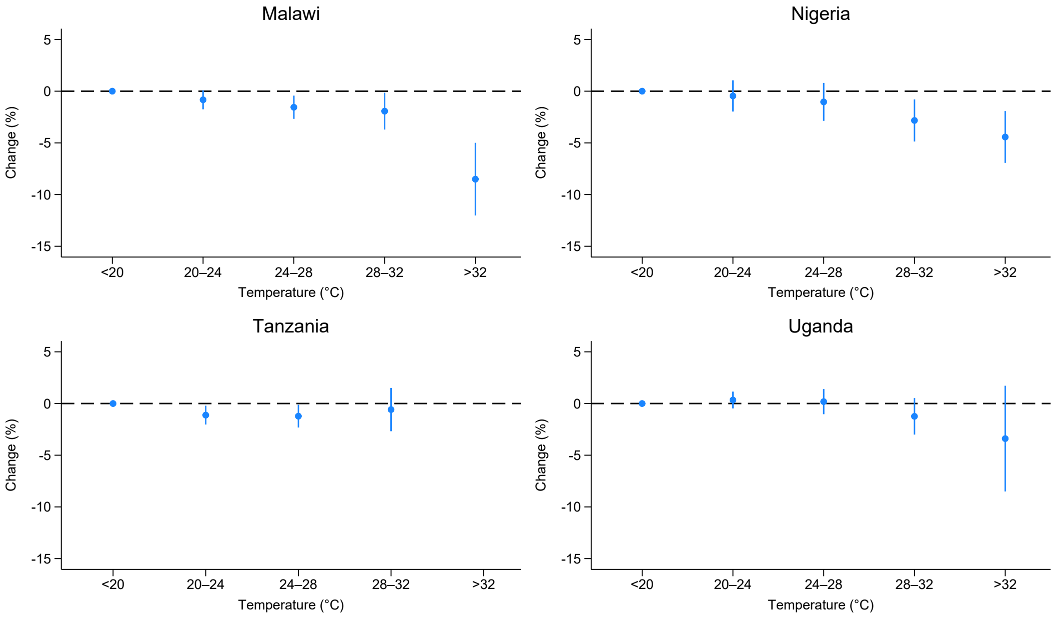

Figure B5. **Growing season temperatures on total yield, by country.** Coefficients are to be interpreted as percentage change. ha: hectare; kg: kilogram. Sample weights are used.
Back to Section 5.1.

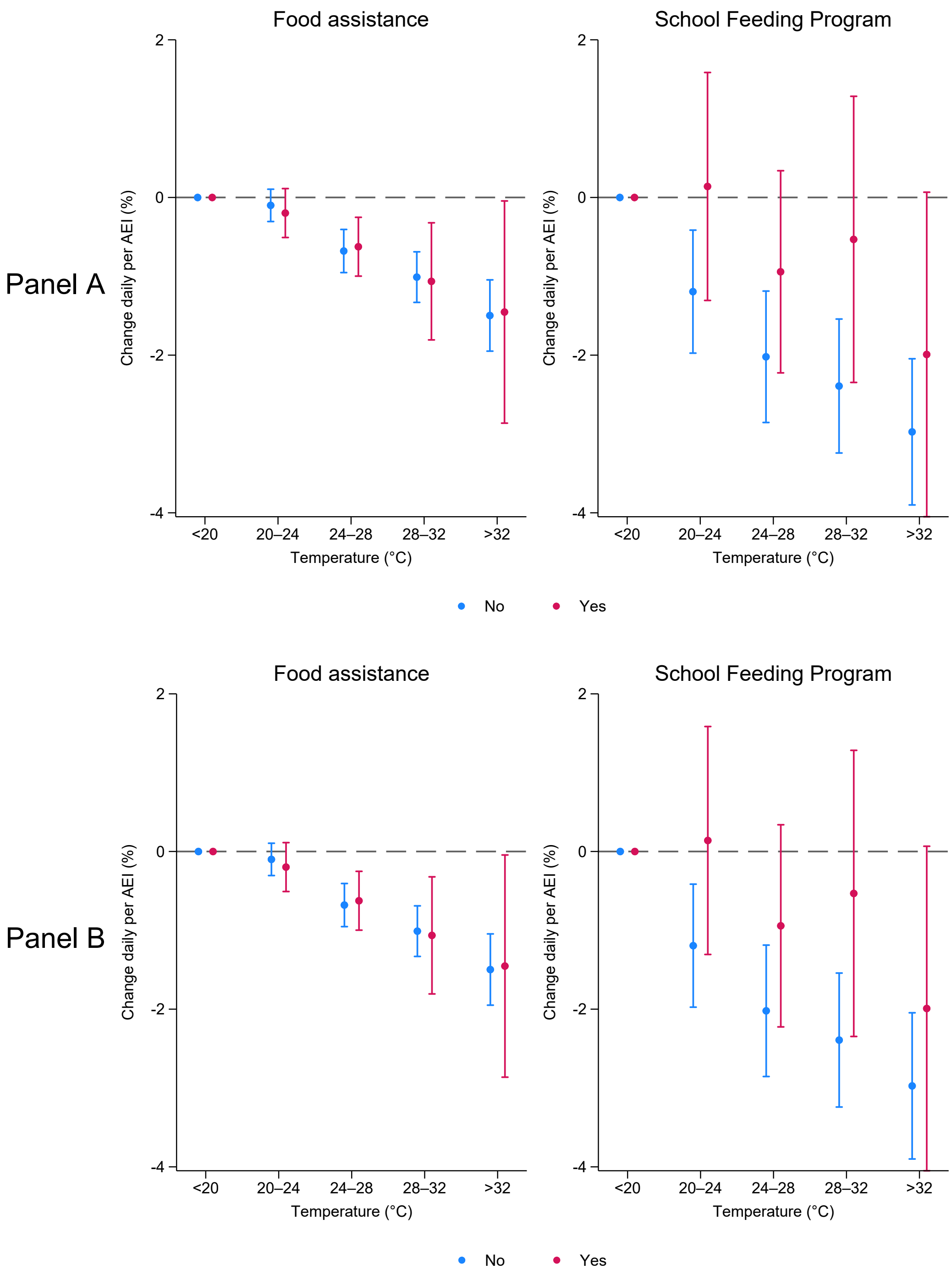

Figure B6. **Last growing season temperatures on energy (Panel A) and protein availability (Panel B), by social protection programme.** Coefficients are to be interpreted as percentage change. Sample weights are used.

Back to Section 6.

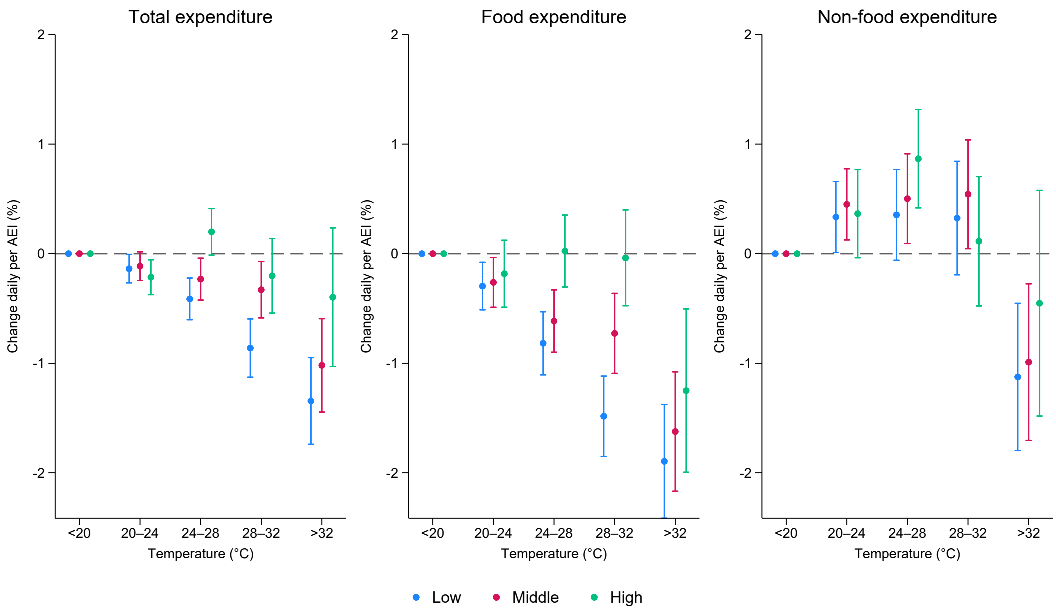

Figure B7. **Last growing season temperatures and expenditure, by household wealth.** Coefficients are to be interpreted as percentage change. Malawi is excluded because data is only available for the first two waves. Disaggregated data on total expenditure between food and non-food items is missing for Nigeria wave 5 and for Uganda. Food expenditure includes a valuation of own-produced food consumption at market prices. All expenditure values are in 2020 USD. Sample weights are used.
Back to Section 5.2.

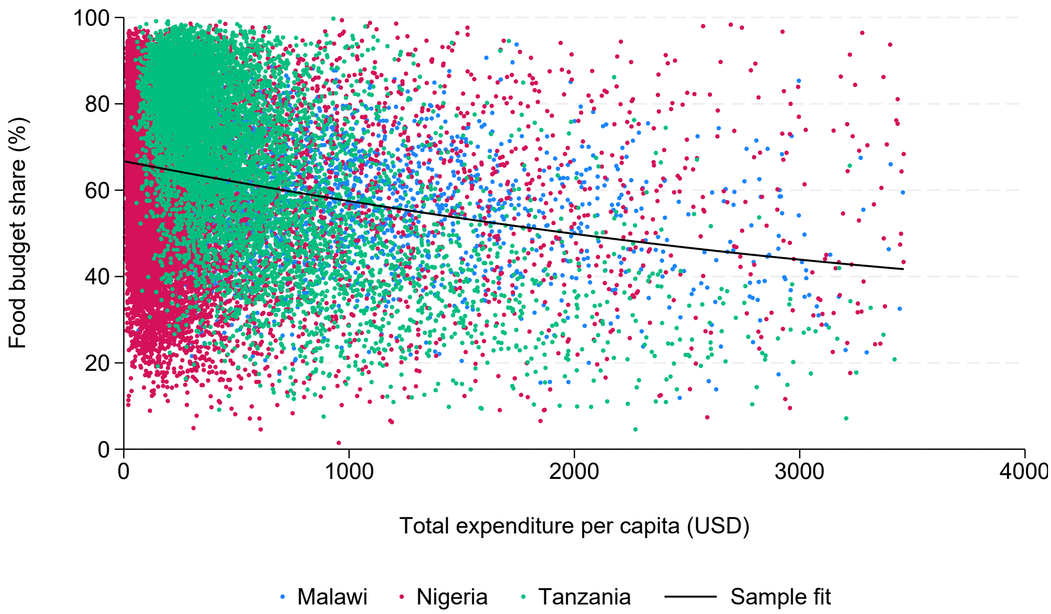

Figure B8. **Food expenditure as a share of total expenditure.** Disaggregated data on total expenditure between food and non-food items is missing for Nigeria wave 5 and for Uganda. Food expenditure includes a valuation of own-produced food consumption at market prices. All expenditure values are in 2020 USD.
Back to Section 5.2.

C. Appendix Tables

	Daily energy per AEI		Adequacy		
			$\geq 100\%$	$\geq 80\%$	$\geq 60\%$
20–24C	-0.283***	-0.274***	-0.002***	-0.002***	-0.001
	(0.088)	(0.088)	(0.001)	(0.001)	(0.001)
24–28C	-0.636***	-0.631***	-0.004***	-0.004***	-0.003***
	(0.119)	(0.120)	(0.001)	(0.001)	(0.001)
28–32C	-0.870***	-0.868***	-0.005***	-0.005***	-0.004***
	(0.143)	(0.144)	(0.001)	(0.001)	(0.001)
$>$ 32C	-1.315***	-1.316***	-0.007***	-0.010***	-0.011***
	(0.212)	(0.212)	(0.002)	(0.002)	(0.002)
Precipitations	0.072*	0.074*	0.001***	0.001*	0.000*
	(0.040)	(0.041)	(0.000)	(0.000)	(0.000)
Precipitations^2	-0.000	-0.000	-0.000**	-0.000*	-0.000*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
N	43151	43151	43151	43151	43151
R^2	0.572	0.572	0.495	0.499	0.497
Household FE	Yes	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes	Yes	Yes
Month-of-year FE	No	Yes	Yes	Yes	Yes
Precipitations	Yes	Yes	Yes	Yes	Yes

Table C1. **Last growing season weather and energy availability and adequacy.** It represents the effect of a 1 percentage point increase in the share of days compared to the reference bin ($<$ 20°C). Columns (1) and (2): coefficients are to be interpreted as percentage change. Columns (3), (4), and (5): coefficients are to be interpreted as percentage point changes. Adequacy represents households above 100%, 80%, and 60% of weekly dietary requirements, estimated for each household based on age and sex composition. Robust clustered standard errors at the household level. Sampling weights are used. AEI: Adult Equivalent Index. FE: fixed effects.
Back to Section 4.1.

20–24C	-0.218***	-0.255**	-0.231**	-0.249**
	(0.060)	(0.109)	(0.105)	(0.111)
24–28C	-0.352***	-0.836***	-0.503***	-0.542***
	(0.075)	(0.143)	(0.141)	(0.148)
28–32C	-0.685***	-1.159***	-0.337*	-0.359*
	(0.097)	(0.168)	(0.196)	(0.203)
< 32C	-0.944***	-1.636***	-0.934***	-0.978***
	(0.157)	(0.233)	(0.324)	(0.331)
N	43151	29807	43146	43119
R^2	0.509	0.590	0.585	0.588
Household FE	Yes	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	No	Yes
Region $\times$ Year FE	No	No	Yes	No
Month-of-year FE	Yes	Yes	Yes	No
Region $\times$ Month-of-year FE	No	No	No	Yes
Precipitations	Yes	Yes	Yes	Yes

Table C2. **Robustness: Last growing season weather and energy availability.** Column (1): Main specification without sampling weights; Column (2): Main specification without Uganda due to bimodal growing season; Column (3): More granular Region $\times$ Year FE; Column (4): More granular Region $\times$ Month-of-Year FE. Sampling weights are used. FE: fixed effects.
Back to Section 4.1.

	HAZ				Stunting			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
l.x	0.0228	0.0003	0.0796	0.0003	-0.0199	-0.0003**	-0.0573***	-0.0004**
	(0.0522)	(0.0005)	(0.0692)	(0.0005)	(0.0161)	(0.0002)	(0.0205)	(0.0001)
l.y	-0.1246***	-0.1245***	-0.1246***	-0.1245***	-0.1131***	-0.1124***	-0.1123***	-0.1123***
	(0.0127)	(0.0127)	(0.0127)	(0.0127)	(0.0108)	(0.0108)	(0.0108)	(0.0108)
y0	0.5897***	0.5899***	0.5894***	0.5897***	0.6876***	0.6878***	0.6863***	0.6870***
	(0.0173)	(0.0172)	(0.0172)	(0.0172)	(0.0130)	(0.0130)	(0.0130)	(0.0130)
N	4779	4779	4779	4779	4779	4779	4779	4779
R^2	0.384	0.384	0.384	0.384	0.432	0.432	0.432	0.432
Mean Y	-1.335	-1.335	-1.335	-1.335	0.292	0.292	0.292	0.292
X variable	100% kcal	ln(kcal_paei)	100% prot	ln(prot_paei)	100% kcal	ln(kcal_paei)	100% prot	ln(prot_paei)
Child controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Correlated RE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table C3. **Mundlak correlated random effects results on the impact of last period energy and protein availability and adequacy on HAZ and stunting.** adq: adequacy; kcal: kilocalorie, paei: per adult equivalent unit, prot: protein. FE: fixed effects.
Back to Section 4.3.

	Total labor	Family labor	Hired labor
20–24C	1.001	1.002	0.994
	(0.002)	(0.002)	(0.004)
24–28C	1.010***	1.012***	0.983***
	(0.004)	(0.004)	(0.005)
28–32C	1.017***	1.020***	0.973***
	(0.005)	(0.005)	(0.006)
> 32C	1.033***	1.037***	0.958***
	(0.007)	(0.007)	(0.008)
N	30589	30379	23973
R^2	0.858	0.852	0.607
Household FE	Yes	Yes	Yes
Country $\times$ Year FE	Yes	Yes	Yes
Precipitations	Yes	Yes	Yes
Model	PPML	PPML	PPML

Table C4. **Growing season temperature on farm labor during the agricultural season.** The PPML specification uses the computationally efficient estimator for Poisson regressions using high-dimensional fixed effects developed by Correia (2016). For PPML, R^2 is a pseudo-R^2 and coefficients are exponentiated and should be interpreted as IRR. It drops separated observations from the estimation sample to avoid problems of perfect fit in likelihood estimations. IRR: Incidence risk ratio; PPML: Poisson pseudo-maximum likelihood.

Back to Section 5.2.